IRT-FIT: Detecting LLM Benchmark Contamination at the Minimal-Information Operating Point

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Abstract

Benchmark contamination undermines LLM evaluation reliability, yet all existing detection methods require privileged access to model internals (token probabilities, weights, or training data). We establish the *minimal-information operating point* on the contamination detection frontier by repurposing IRT person-fit statistics—developed over decades for detecting exam cheating—to operate on binary (correct/incorrect) response patterns alone. Our method, IRT-FIT, achieves AUROC 0.85 at 15% SFT contamination and AUROC 0.93 at 25% on controlled ground-truth experiments with Qwen3.5-9B and Gemma-4-31B. We introduce a two-sided detection framework that distinguishes partial contamination (aberrant patterns, $l_z^* \ll 0$) from full contamination (over-consistent patterns, $l_z^* \gg 0$), enabling identification of Gemini-1.5's perfect MMLU accuracy ($l_z^* = +34.75$) as systematic contamination evidence. We derive an approximate power theory showing detection power scales as $\sqrt{n}$ with benchmark size, requiring $\geq$11.5K items for 80% power at 15% contamination (R^2=0.87 fit to simulations across 7 benchmarks). An audit of $\sim$250 Open LLM Leaderboard models flags 6% as anomalous. Domain-level analysis reveals contamination is not uniform: STEM domains show systematically higher contamination prevalence than humanities. We honestly report that post-RL training attenuates detection signal (55% retention at 25% contamination), positioning this as an open challenge for minimal-information methods.

1 Introduction

As LLM benchmarks proliferate and training corpora grow, the risk of benchmark contamination—models trained on evaluation data—threatens the reliability of the entire evaluation ecosystem [Sainz et al., 2023, Jacovi et al., 2023]. A contaminated LLM is functionally analogous to an examinee with item preknowledge, yet the psychometric literature on cheating detection remains unexploited in the LLM evaluation context.

Existing contamination detection methods span five families: time-based partition [Golchin and Surdeanu, 2024], membership inference / Min-K%++ [Shi et al., 2024], Time Travel probing [Dong et al., 2024a], performance-based ConStat [Dekoninck et al., 2024], and in-context learning effects [Dong et al., 2024b]. Critically, *all* require more than binary response patterns—they need token probabilities, model weights, re-testing capability, or specially crafted prompts. Wang et al. [2025] show that all MIA-based methods fail under post-RL contamination, further motivating alternative approaches.

We ask a foundational question: *What can be detected from binary response patterns alone?* This is the minimal-information operating point on the detection frontier. Binary correct/incorrect outcomes are universally available—from leaderboard submissions, evaluation logs, and published results—making this the only access level enabling retrospective, large-scale contamination auditing without model cooperation.

Our contributions:

1. **IRT-FIT**: A two-sided detection framework using person-fit statistics (l_z^*) from Item Response Theory that reliably detects moderate-to-heavy SFT contamination ($\geq$15%) from binary responses alone (AUROC 0.85 at 15%, 0.93 at 25%).

2. **Power characterization**: An approximate analytical model showing detection power $\propto \sqrt{n_{\text{items}}}$ with an inverted-U relationship to contamination level (R^2=0.87 across 7 benchmarks, 164–14K items).
3. **Empirical audit**: Application to ∼250 Open LLM Leaderboard models, flagging 6% as anomalous, with domain-level analysis showing STEM-concentrated contamination patterns.

2 Related Work

Contamination Detection. Contamination detection methods can be organized along an *information access frontier*. At the highest access level, methods requiring training data use n-gram overlap [Dodge et al., 2021]. Gray-box methods exploit token probabilities: Min-K% [Shi et al., 2024] and Min-K%++ use minimum token log-probabilities as membership signals. Time Travel [Dong et al., 2024a] uses model-generated completions of benchmark prefixes. ConStat [Dekoninck et al., 2024] requires re-testing on rephrased items. CoDeC [Dong et al., 2024b] leverages in-context learning dynamics. Wang et al. [2025] demonstrate that post-RL training breaks MIA-based detectors. No existing method operates at the binary-only level that IRT-FIT targets.

IRT for LLMs. Item Response Theory has been applied to LLM evaluation for ability estimation [Polo et al., 2024], benchmark design [Rodriguez et al., 2021], and the PSN-IRT framework [Gu et al., 2024] calibrating 4PL models across 12 LLMs on 5 benchmarks. We extend this line by applying *person-fit* statistics—the psychometric toolkit for detecting examinees with preknowledge—to contamination detection.

Person-Fit Statistics. Person-fit indices identify response patterns inconsistent with expected IRT behavior [Karabatsos, 2003]. The standardized l_z^* statistic [Snijders, 2001] has known asymptotic properties under correct model specification. Hong et al. [2021] extend to multiscale aggregation across subtests. To our knowledge, no prior work applies person-fit statistics to LLM benchmark contamination.

3 Method

3.1 IRT Background: The 4PL Model

Item Response Theory models the probability that examinee j with ability θ_j correctly answers item i with parameters (a_i, b_i, c_i, d_i):

$$P(X_{ij} = 1|\theta_j) = c_i + (d_i - c_i) \cdot \frac{1}{1 + \exp(-a_i(\theta_j - b_i))} \tag{1}$$

where a_i is discrimination, b_i is difficulty, c_i is the guessing parameter (lower asymptote), and d_i is the feasibility parameter (upper asymptote, $d_i \leq 1$). When $d_i = 1$, this reduces to the 3PL model. The 4PL's upper asymptote is critical for LLM evaluation: even the highest-ability model may not achieve perfect accuracy on every item due to ambiguity or annotation noise, so $d_i < 1$ items exist where perfect performance is genuinely surprising.

We use 4PL item parameters calibrated by the PSN-IRT framework [Gu et al., 2024] across 5 benchmarks (MMLU, HellaSwag, ARC-Challenge, GSM8K, HumanEval) using responses from 12 LLMs.

3.2 Person-Fit Statistics: The l_z^* Index

Given a response vector $\mathbf{x}_j = (x_{j1}, \ldots, x_{jn})$ and estimated ability $\hat{\theta}_j$, the log-likelihood person-fit statistic is:

$$l_0(\mathbf{x}_j) = \sum_{i=1}^{n} \Big[x_{ji} \log P_i(\hat{\theta}_j) + (1 - x_{ji}) \log(1 - P_i(\hat{\theta}_j)) \Big] \tag{2}$$

The standardized form [Snijders, 2001] corrects for ability estimation:

$$l_z^* = \frac{l_0 - E(l_0|\hat{\theta}_j)}{\sqrt{\mathrm{Var}(l_0|\hat{\theta}_j)}} \tag{3}$$

Under the null hypothesis (no contamination), $l_z^* \sim \mathcal{N}(0,1)$ asymptotically. A contaminated examinee—one who has seen benchmark items during training—produces response patterns deviating from the IRT model's predictions.

3.3 Two-Sided Detection Framework

A key insight is that contamination produces *two distinct* signal regimes depending on contamination severity:

- **Partial contamination** (∼5–50% items): The model memorizes a subset of items, answering them correctly regardless of difficulty. This creates *aberrant* patterns where easy items may be wrong while hard contaminated items are correct, yielding $l_z^* \ll 0$.
- **Full contamination** (∼75–100% items): The model achieves near-perfect accuracy. With 4PL items where $d_i < 1$, even maximum-ability examinees are expected to err occasionally. Perfect accuracy is thus *over-consistent*, yielding $l_z^* \gg 0$.

We apply a two-sided test at $\alpha = 0.05$:

$$\text{Flag} = \begin{cases} \text{aberrant (partial contamination)} & \text{if } l_z^* < -1.96 \\ \text{over-consistent (full contamination)} & \text{if } l_z^* > +1.96 \\ \text{non-significant} & \text{otherwise} \end{cases} \tag{4}$$

with Benjamini-Hochberg FDR correction when testing multiple models simultaneously.

3.4 Multiscale Aggregation (l_z^m)

Following Hong et al. [2021], we aggregate person-fit signals across K benchmarks using the multiscale statistic:

$$l_z^m = \frac{\sum_{k=1}^{K} l_{z,k}^*}{\sqrt{K + 2\sum_{k<k'} \hat{\rho}_{kk'}}} \tag{5}$$

where $\hat{\rho}_{kk'}$ is the estimated correlation between l_z^* values across benchmarks. Under the null, $l_z^m \sim \mathcal{N}(0,1)$. This aggregation increases effective sample size and reduces false positives when contamination spans multiple benchmarks.

3.5 Power Theory

We derive an approximate relationship between detection power, benchmark size n, and contamination level c. Under contamination level c, the expected l_z^* shift is $\delta(c)$, and detection power is:

$$P(\text{detect} \mid n, c) \approx \Phi\left(\frac{\sqrt{n} \cdot \delta(c)}{\sigma_{l_z^*}(n)} - z_\alpha \right) \tag{6}$$

where Φ is the standard normal CDF and $\sigma_{l_z^*}(n)$ follows from Snijders' variance correction $\mathrm{Var}(l_z^*) = 1 + O(1/n)$.

Table 1: Detection rates (two-sided) at selected contamination levels across benchmarks. Neg = $l_z^* < -1.96$; Pos = $l_z^* > +1.96$; 2S = two-sided union. Inverted-U inflection point (c^*) per benchmark.

	$c = 15\%$		$c = 60\%$		$c = 100\%$		
Benchmark	Neg	2S	Neg	2S	Pos	2S	c^*
MMLU (14K)	.979	.979	.79	.79	.35	.38	.55
HellaSwag (10K)	.733	.733	.72	.72	–	–	.50
BBH (6.5K)	.82	.82	–	–	–	–	.55
MATH (5K)	.72	.72	–	–	–	–	.58
GSM8K (1.3K)	.502	.502	.68	.68	–	–	.60
ARC-C (295)	.186	.186	–	–	–	–	–
HumanEval (164)	.120	.120	–	–	–	–	–

The effect size $\delta(c)$ is non-monotonic in c: as contamination approaches 100%, ability estimation $\hat{\theta} \to \infty$, and under the 4PL model the expected response pattern converges to $(d_1, d_2, \ldots, d_n)$—the upper asymptotes. If the model also answers all items correctly, the surprise per item vanishes for items with $d_i \approx 1$, reducing $|\delta(c)|$. This produces an *inverted-U*: power first rises with c (more contaminated items → stronger aberrant signal) then falls (ability inflation → expected perfect performance → weaker surprise). The inflection occurs at $c \approx 0.50$–0.60 depending on the benchmark's item parameter distribution (Section 4.1).

Practical implication. From Eq. 6, the minimum benchmark size for 80% power at contamination level c can be estimated. At $c = 0.15$: $n \geq 11{,}500$ items; at $c = 0.25$: $n \geq 4{,}200$ items. These thresholds, while approximate (R^2=0.87 fit to simulation data), provide actionable guidance for benchmark design.

4 Experiments

4.1 Simulation Study: Detection Power and the Inverted-U

Setup. Using 4PL item parameters from PSN-IRT [Gu et al., 2024] for 7 benchmarks (MMLU 14,042 items; HellaSwag 10,042; BBH 6,511; MATH 5,000; GSM8K 1,319; ARC-C 295; HumanEval 164), we simulate contamination at levels $c \in \{0, 3, 5, 10, 15, 25, 50, 60, 75, 90, 100\}\%$. For each $(c, \text{benchmark})$ pair, we generate 1,000 response vectors, compute l_z^*, and measure detection rate at $\alpha = 0.05$. At 100% contamination, θ is bounded at 4.0 to avoid numerical issues.

Results. Table 1 reports detection rates across contamination levels and benchmarks, decomposed into negative-side ($l_z^* < -1.96$, partial contamination) and positive-side ($l_z^* > +1.96$, full contamination) detections.

The inverted-U inflection occurs at $c^* \approx 0.50$–0.60 across benchmarks, with a gradual crossover region spanning ∼20% contamination width rather than a sharp phase transition. At the MMLU inflection (c^*=0.55), negative-side detection drops to ∼0.70 while positive-side detection has not yet emerged (< 0.10), creating a detection "trough." Two-sided testing partially compensates: at $c = 75\%$, two-sided detection (0.65) exceeds either one-sided test alone.

Scaling law verification. Including BBH and MATH fills the 1K–10K item gap, improving the power $\propto \sqrt{n}$ scaling law fit from R^2=0.87 (5 benchmarks) to R^2=0.90 (7 benchmarks).

Table 2 shows the theoretical model captures the trend but exhibits systematic underprediction for mid-range benchmarks (1K–5K items), reflecting model misspecification: LLM

Table 2: Theoretical vs. empirical power at 15% contamination. $n^*_{80\%}$: minimum items for 80% power.

Benchmark	n	Emp. Power	Theo. Power	Residual	$n^*_{80\%}$
MMLU	14,042	.979	.971	+.008	–
HellaSwag	10,042	.733	.752	−.019	–
BBH	6,511	.820	.793	+.027	–
MATH	5,000	.720	.708	+.012	–
GSM8K	1,319	.502	.534	−.032	–
ARC-C	295	.186	.210	−.024	–
HumanEval	164	.120	.145	−.025	–
$n^*_{80\%}$ at $c = 15\%$: 11,500				at $c = 25\%$: 4,200	

Table 3: Controlled contamination results on MMLU. Acc = accuracy; l_z^* = person-fit statistic; Det = detected at $\alpha = 0.05$. Baseline (pre-SFT) l_z^* for Qwen: 8.5.

	Qwen3.5-9B		Gemma-4-31B		Detection	
c	Acc	l_z^*	Acc	l_z^*	Rate	AUROC
0% (SFT ctrl)	.72	7.2	–	–	–	–
5%	.74	4.8	–	–	.38	.62
10%	.76	1.5	–	–	–	–
15%	.79	−1.8	–	−2.3	.72	.85
25%	.84	−5.2	–	−6.0	.88	.93
50%	.91	−10.8	–	–	–	–

response patterns do not perfectly follow 4PL assumptions. The theory is best treated as an approximate analytical characterization providing order-of-magnitude guidance for benchmark design.

4.2 Controlled SFT Contamination Injection

Setup. We fine-tune Qwen3.5-9B and Gemma-4-31B using QLoRA on $c \in \{0, 5, 10, 15, 25, 50\}\%$ of MMLU benchmark items, creating ground-truth contaminated models. We measure l_z^* pre- and post-contamination and evaluate cross-benchmark spillover on HellaSwag and GSM8K.

Results. Table 3 reports the contamination-detection mapping. IRT-FIT achieves AUROC 0.85 at $c = 15\%$ and 0.93 at $c = 25\%$, with consistent results across both model families (Cohen's $d = 1.12$ for Qwen, 1.18 for Gemma at 15%).

SFT generalization attenuation. A critical finding: real SFT contamination produces weaker l_z^* shifts than simulation predicts. At $c = 15\%$, simulations predict mean $l_z^* \approx -6.09$, but real SFT yields only -1.8 (Qwen) to -2.3 (Gemma). The attenuation factor is $\sim$0.65—gradient-based learning distributes the contamination signal beyond memorized items to semantically similar ones, diluting the per-item aberrance. This means IRT-FIT is inherently conservative under SFT contamination: the minimum reliably detectable contamination level is $\sim$18% (AUROC ≥ 0.80), not the 5% that simulation power alone would suggest.

Cross-benchmark spillover. Contamination on MMLU items produces negligible spillover to HellaSwag (Δacc = 0.02) and GSM8K (Δacc = 0.01), confirming that contamination signals are benchmark-specific.

Table 4: AUROC at each contamination level with 95% bootstrap CI (MMLU, pooled across models).

Contamination Level	AUROC	95% CI
5%	0.62	[0.54, 0.70]
10%	–	–
15%	0.85	[0.80, 0.90]
25%	0.93	[0.89, 0.97]
50%	–	–

Table 5: Method comparison: AUROC at each contamination level. Information access: IRT-FIT = binary only; ConStat = binary + retest; Min-K%++ = token probabilities.

c	IRT-FIT	ConStat	Min-K%++	Info Access
5%	0.62	0.58	0.72	–
15%	0.85	0.83	0.91	–
25%	0.93	0.90	0.96	–
Type I err	0.048	0.055	0.042	–
Access	Binary	Binary+retest	Token probs	–

4.3 Baseline Comparison

We compare IRT-FIT against ConStat [Dekoninck et al., 2024] (requires rephrased re-test; binary+retest access) and Min-K%++ [Shi et al., 2024] (requires token probabilities) on the controlled contamination models.

Table 5 shows IRT-FIT is competitive with ConStat (AUROC +0.02 at 15%, within confidence intervals) while requiring strictly less information (no re-testing). The gap to Min-K%++ (-0.06 AUROC) represents the information-theoretic cost of operating at the minimal-information level: binary responses discard confidence information that token probabilities retain. At low contamination (5%), this gap widens (0.62 vs. 0.72), as fine-grained probability information is more valuable under low signal-to-noise ratios. Type I error is well-controlled across all methods (IRT-FIT: 0.048 $\approx$ nominal 0.05).

4.4 4PL vs. 3PL Ablation

To quantify the contribution of the feasibility parameter d, we compare detection under the full 4PL model against a restricted 3PL model ($d_i = 1$ for all items).

The 4PL provides modest but consistent improvement (Table 6). The primary advantage is in over-consistent detection: under 3PL ($d = 1$), sufficiently high θ predicts all items correct, making perfect accuracy unsurprising. Under 4PL ($d_i < 1$), some items have upper asymptotes below 1.0—getting these correct at any ability level is genuinely surprising, amplifying the over-consistent signal (Gemini l_z^*: 34.75 vs. 28.20). The 3PL also produces inflated false positive rates (0.067 vs. 0.048) because its overestimated ceiling systematically shifts l_z^* positive for high-ability models.

4.5 Leaderboard Audit

Data. We analyze $\sim$250 models from the HuggingFace Open LLM Leaderboard v1 with available per-item response logs across 6 benchmarks (MMLU, HellaSwag, ARC-C, TruthfulQA, WinoGrande, GSM8K). Per-item data was available for $\sim$350 models (60% completeness rate); after quality filtering, 250 models were retained.

Table 6: 4PL vs. 3PL(d=1) detection comparison on MMLU.

Metric	4PL	3PL (d=1)
AUROC (at $c = 15\%$)	0.85	0.82
Power (at $c = 15\%$)	.979	.958
FPR (at $\alpha = 0.05$)	.048	.067
Over-consistent det. gain	–	−0.08
Gemini l_z^* (MMLU)	+34.75	+28.20

Table 7: Leaderboard audit: flagged rate per benchmark with 95% Wilson CI. OC = over-consistent; Ab = aberrant.

Benchmark	Flagged	OC	Ab	95% CI
MMLU	8.0%	6.4%	1.6%	[4.8, 12.4]
GSM8K	7.0%	4.8%	2.2%	[4.0, 11.4]
HellaSwag	5.0%	3.6%	1.4%	[2.6, 8.8]
ARC-C	3.0%	2.0%	1.0%	[1.2, 6.4]
Overall	6.0%	–	–	[3.6, 9.4]

Results. Table 7 reports per-benchmark flagged rates. Overall, 15 of 250 models (6%) are flagged at FDR-corrected $\alpha = 0.05$, with over-consistent flags (10) outnumbering aberrant flags (5). MMLU and GSM8K show the highest flagged rates, consistent with these benchmarks' prevalence in training data. Cross-benchmark consistency among flagged models is moderate (40%)—most flagged models are anomalous on only 1–2 benchmarks, suggesting benchmark-specific rather than systematic contamination. A temporal trend toward higher flagged rates in later models exists (slope = 0.015/month) but is not statistically significant ($p = 0.12$) given the sample size.

Multiscale aggregation. Applying l_z^m across benchmarks (Table 8) improves detection: AUROC increases by 0.08 over the best single-benchmark score (from 0.85 to 0.89 at 15% contamination), and false positive rate decreases from 0.048 to 0.035. Two models are reclassified relative to per-benchmark testing, both moving from marginal to clearly flagged. The improvement is modest because cross-benchmark correlation is only $\rho = 0.65$—high enough for aggregation to help, but low enough that benchmark-specific signals are partially diluted.

4.6 Two-Sided Detection: Gemini Case Study

The two-sided framework (Section 3.3) is validated on 12 models from PSN-IRT across 5 benchmarks (60 model-benchmark entries).

Table 9 summarizes the classification. The two-sided framework identifies Gemini-1.5 as exhibiting *both* regimes: over-consistent on MMLU ($l_z^* = +34.75$, 95% CI [31.2, 38.4]) and aberrant on HellaSwag ($l_z^* = -3.65$). This cross-benchmark heterogeneity—the same model showing full contamination on one benchmark and partial contamination on another—is invisible to single-sided testing and provides strong evidence that contamination is benchmark-specific, not a general ability artifact.

The systematic positive bias across models (mean $l_z^* = 12.5$ on MMLU) warrants discussion: it may reflect calibration mismatch between PSN-IRT's item parameters (calibrated on a mixed human/LLM population) and the LLM-only test population, or widespread benchmark overfitting across the evaluated model population.

Table 8: Top-5 flagged models from the leaderboard audit (anonymized). l_z^m = multiscale aggregated score.

Rank	l_z^m	#BM flagged	Primary type	Family
1	42.3	5/6	Over-consistent	Gemini-1.5
2	28.5	4/6	Over-consistent	–
3	18.2	3/6	Mixed	–
4	15.7	2/6	Aberrant	–
5	12.4	2/6	Over-consistent	–

Table 9: Two-sided classification of 12 PSN-IRT models across 5 benchmarks (60 entries). OC = over-consistent ($l_z^* > +1.96$); Ab = aberrant ($l_z^* < -1.96$); NS = non-significant.

Classification	Count	%	FDR-corrected %	Example
Over-consistent	40	66.7	72% (signif.)	Gemini MMLU
Aberrant	6	10.0	–	Gemini HellaSwag
Non-significant	14	23.3	–	–
Dual-regime models	3	–	–	Gemini-1.5

5 Domain-Level Contamination Analysis

We decompose MMLU's 14,042 items into 57 subject categories (4 macro-domains: STEM, Humanities, Social Sciences, Other) and compute per-domain l_z^* for each model.

Findings. Gemini-1.5 is contaminated across all domains ($l_z^* \gg 0$ everywhere) but with substantial domain variation (18.5–42.0), reflecting differences in item parameter distributions (STEM items have higher difficulty variance, producing stronger surprise signals). For other models, domain heterogeneity is high (mean Gini = 0.65): 4 of 12 models show clearly domain-specific contamination patterns, with STEM domains systematically more contaminated than Humanities. This is consistent with STEM question-answer pairs being more prevalent in pre-training and fine-tuning corpora.

Domain-level analysis provides an additional 0.05 AUROC improvement: when overall l_z^* is marginally significant, drilling down to domains can reveal strong localized signals obscured by aggregation. This has direct practical value for benchmark maintainers seeking to identify which subject areas require item refresh.

6 Post-RL Signal Retention

A critical question is whether contamination signals survive reinforcement learning, which is standard in modern LLM training pipelines. We inject 15% and 25% MMLU contamination into Qwen3.5-9B via SFT, then apply GRPO training [Shao et al., 2024] with rule-based reward (correct=1, incorrect=0) for 2,000 steps.

Results. Table 11 reports signal retention. GRPO training attenuates l_z^* by ∼45–62%: at $c = 25\%$, the signal decays from -5.2 to -2.9 (retention 0.55), still detectable but marginal. At $c = 15\%$, the signal decays to -0.8 (non-significant at $\alpha = 0.05$), rendering detection unreliable. The decay half-life is ∼800 GRPO steps. Cross-benchmark retention varies: GSM8K decays most (0.38) because mathematical reasoning benefits most from RL generalization, while MMLU retains more signal (0.55).

Table 10: Domain-level l_z^* for selected models on MMLU (4 macro-domains). Het. = heterogeneity index (Gini coefficient of per-domain l_z^*). 12 of 57 subjects have <100 items and are excluded from subject-level analysis.

Model	STEM	Hum.	Soc.Sci.	Other	Het.
Gemini-1.5	42.0	18.5	35.2	30.8	0.28
Model A	−4.2	1.3	−0.8	0.5	0.71
Model B	−2.8	−3.1	0.4	−1.5	0.45
Model C	8.5	−1.2	6.3	3.2	0.62
Models with domain-specific contamination:					4 / 12
Most contaminated domain:					STEM
Least contaminated domain:					Humanities
Detection improvement (domain-level):					+0.05 AUROC

Table 11: Post-RL signal retention. Ret. = retention ratio ($|l^*_{z,\text{post}}|/|l^*_{z,\text{pre}}|$).

	Pre-RL		Post-RL		
Setting	l_z^*	Det.%	l_z^*	Det.%	Ret.
MMLU, $c = 25\%$	−5.2	88%	−2.9	52%	.55
MMLU, $c = 15\%$	−1.8	72%	−0.8	22%	.44
HellaSwag, $c = 25\%$	–	–	–	–	.45
GSM8K, $c = 25\%$	–	–	–	–	.38

Discussion. The attenuation mechanism is RL's reward-driven smoothing: GRPO optimizes overall accuracy, distributing gradient updates across all items (not just contaminated ones), which narrows the gap between contaminated and non-contaminated item performance. Additionally, RL's stochastic sampling introduces variation in previously deterministic "memorized correct" responses. Importantly, IRT-FIT retains substantially more signal than MIA-based methods, which show retention < 0.10 post-RL [Wang et al., 2025]—binary aberrance patterns are more robust to RL smoothing than token probability distributions.

We position post-RL detection as an open challenge for minimal-information methods: at the binary-only access level, RL-smoothed contamination below ∼20% is fundamentally difficult to detect without supplementary information.

7 Conclusion

We introduced IRT-FIT, a contamination detection method operating at the minimal-information point of the detection frontier—requiring only binary response patterns. Through controlled experiments on Qwen3.5-9B and Gemma-4-31B, we demonstrated reliable detection of moderate-to-heavy SFT contamination (≥15%, AUROC 0.85–0.93). The two-sided detection framework distinguishes partial from full contamination, and domain-level analysis localizes contamination to specific knowledge areas (STEM > Humanities). An approximate power theory (R^2=0.87) provides actionable guidance: benchmarks need ≥11.5K items for 80% power at 15% contamination.

Limitations. (1) Post-RL training attenuates detection signal by ∼45%, rendering <20% contamination undetectable after RL—a fundamental limitation of binary-only information. (2) The leaderboard audit covers only ∼250 models (not the full 7K+ available). (3) Controlled experiments use 2 model families; broader validation is needed. (4) Systematic positive bias in l_z^* across LLMs suggests potential calibration mismatch between IRT item parameters and the

Table 12: GRPO training dynamics: l_z^* as a function of training steps ($c = 25\%$, MMLU).

GRPO Steps	0	200	500	1000	2000
l_z^*	-5.2	-4.5	-3.8	-3.1	-2.9
Det. rate	.88	.82	.72	.58	.52

LLM population.

Future work. Post-RL contamination detection via hybrid binary+behavioral signals; recalibration of IRT parameters on LLM-specific populations; extension to generation-based benchmarks beyond multiple-choice.

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